

Introduction to Earth System

■ Atmosphere

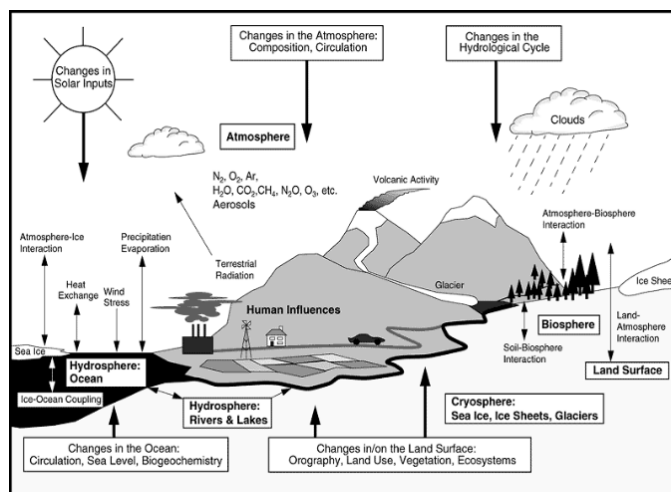
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1

1

Components of the Earth System

- Geosphere
- Atmosphere
- Hydrosphere
- Biosphere

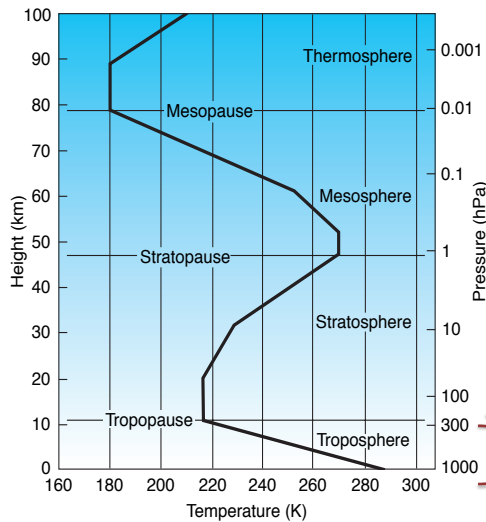


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IPCC, 2001

2

The Atmosphere (by temperature)



A typical midlatitude vertical temperature profile, as represented by the U.S. Standard Atmosphere.

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Stratosphere: Increase of T with height

- Vertical mixing strongly inhibited (ideal for jet planes)
- Residence times of particles: longer
- Air: extremely dry and ozone rich
- Stratopause: max T at 50km ← heating due to absorption of ultraviolet radiation
- Weather balloons
- 99% of the atm's mass (tropo + strato)

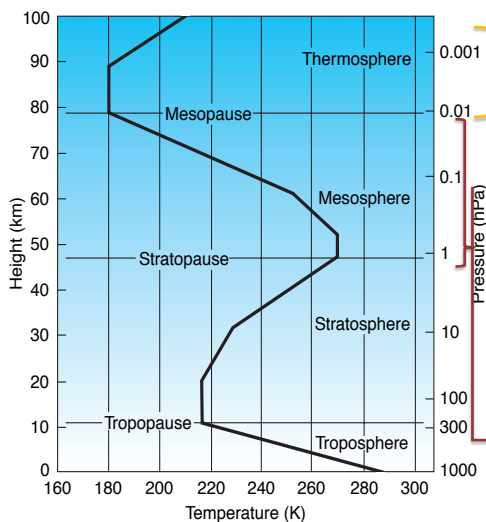
Troposphere: Generally decreasing temperature with height; average lapse rate:

$$\Gamma \equiv \frac{\partial T}{\partial z} \sim 6.5^\circ \text{C km}^{-1}$$

- Height ~18 (tropics) to ~6km (winter pole)
- 80% of the mass of the atm. → prove?
- Almost all weather occurs here
- Temperature inversion exists

3

The Atmosphere (by temperature)



A typical midlatitude vertical temperature profile, as represented by the U.S. Standard Atmosphere.

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Thermosphere: T increase with height

- T can be ~1700°C (absorption of solar radiation & photodissociation (N₂, O₂))
- Mean free pathway ~km
- The amount of energy absorbed is small, but the density is extremely low → large increase in temperature
- **Aurora**

Mesosphere: T decreases with height to a minimum that defines mesopause

- The cold layer: very little ozone (~1% atm ozone) to absorb solar radiation
- Not easy to study (hard to make measurements)
- Mesopause: the coldest place on earth

4



<https://images.wallpapersden.com/>

<https://cdn.wallpapersafari.com/>

<http://geoblog.altervista.org/>

- **Aurora:** when atm gases are excited they emit visible radiation.
- N₂ emits **violet (427.8 nm)**
- O₂ emits **green (557.7 nm)** and **red (630 nm)** light

Q: Why are auroras usually seen near the poles?

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5

The escape zone

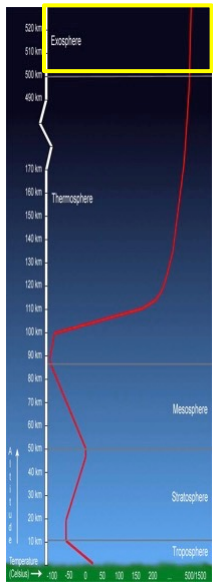
extends until it merges with space

upper limit of our atmosphere

comprised of mainly hydrogen and helium (extremely low densities)

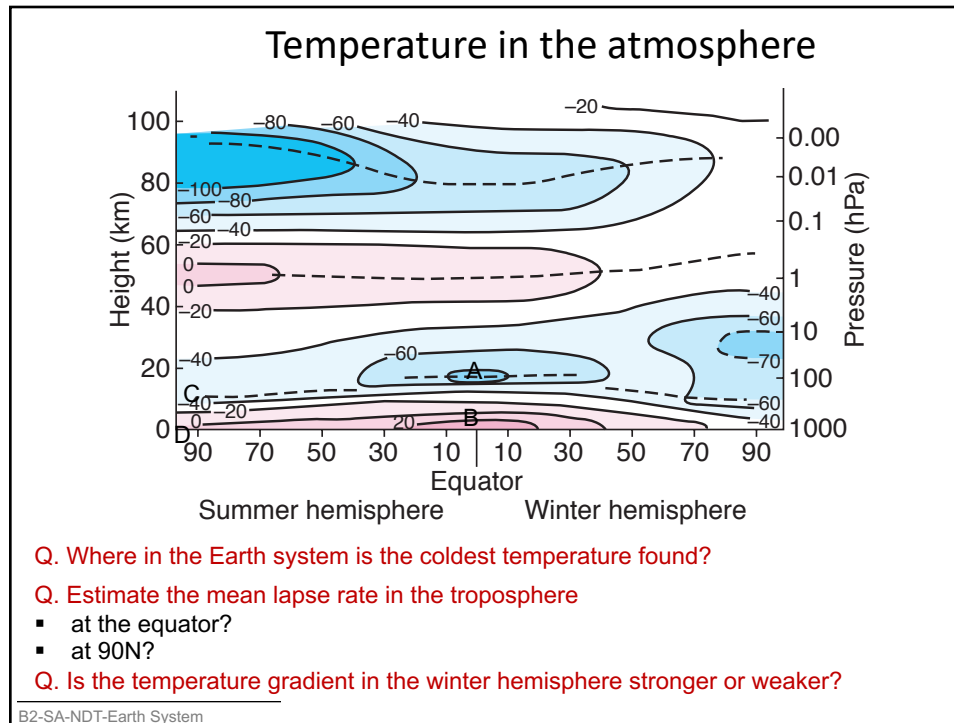
satellites orbit earth in this layer

Exosphere
600+ km



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6



7

The Atmosphere: composition

- The **homosphere** and **heterosphere** are defined by whether the atmospheric gases are well mixed.

Homosphere Lower atmosphere The air is well mixed: ▪ 78.09%: Nitrogen (N_2); ▪ 20.95%: Oxygen (O_2); ▪ 0.93% Argon ▪ & other “trace” gases ▪ CO_2 ~400 ppm (0.037%) ▪ Methan CH_4 ~1.8ppm ▪ Ozone O_3 ~0.1 ppm (90% between 20-50km) ▪ Water (H_2O): various amount, on average around 1% at sea level, and 0.4% over the entire atmosphere	Heterosphere upper atmosphere ▪ The air is not well mixed ▪ Atoms get sorted by atomic weight (by gravity) ▪ H_2 and He can sometimes escape gravity
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8

The Atmosphere (by composition)

Greenhouse Gases

- N_2 and O_2 are transparent to both incoming solar radiation and OLR
→ do not play a role in establishing atmospheric temperature
- **Gases that absorb OLR are called greenhouse gases**

Q. What are the main GHGs?

1. water vapor (H_2O)
2. carbon dioxide (CO_2)
3. methane (CH_4)
4. nitrous oxide (N_2O)
5. ozone (O_3)
6. CFCs

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9

9

Important atmospheric greenhouse gases (2012)

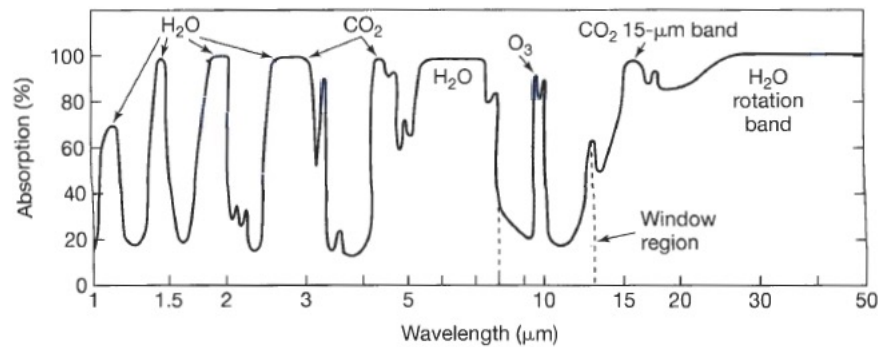
Name	Concentration (ppm by volume)
Water vapor, H_2O	0.1 (South Pole) - 40000 (tropics)
Carbon dioxide, CO_2	370 (427 as of Feb 2025)
Methane, CH_4	1.7 (1.94 as of Nov 2023)
Nitrous oxide, N_2O	0.3
Ozone, O_3	0.01 (at the surface)
Freon-11, CCl_3F	0.00026
Freon-12, CCl_2F_2	0.00054

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10

10

Percentage of radiation absorbed during vertical passage through the atmosphere



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11

11

Are GHGs increasing?

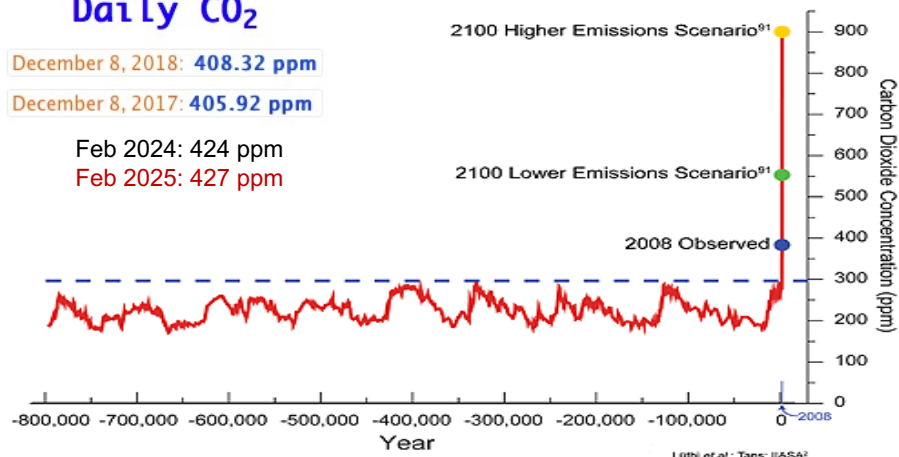
Daily CO₂

December 8, 2018: 408.32 ppm

December 8, 2017: 405.92 ppm

Feb 2024: 424 ppm

Feb 2025: 427 ppm

Lüthi et al.; Tans; NASA²

At 2 ppm rise per year

→ increasing rate of ~80 times that of the fastest natural rate and almost 2000 times the average rate over the past hundreds of thousands of years

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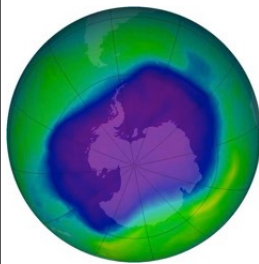
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12

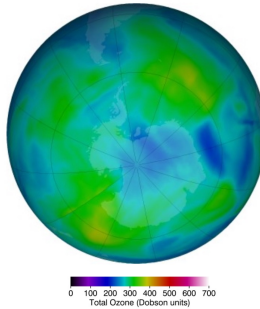
CFCs, HCFCs

- CFCs & HCFCs → man-made GHG
- Montreal Protocol (1989, 196 countries)

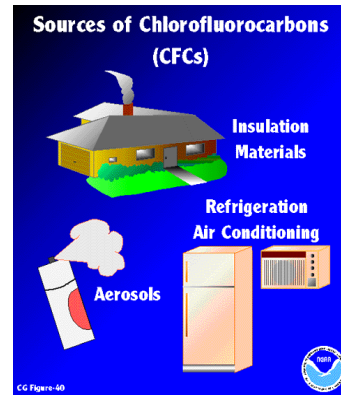
25 March 2024



Ozone hole in Antarctic



0 100 200 300 400 500 600 700
Total Ozone (Dobson units)



Kofi Annan "perhaps the single most successful international agreement to date has been the Montreal Protocol"

13

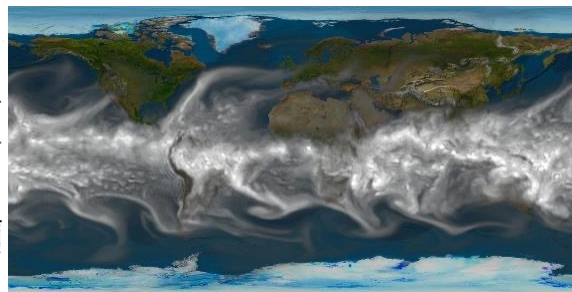
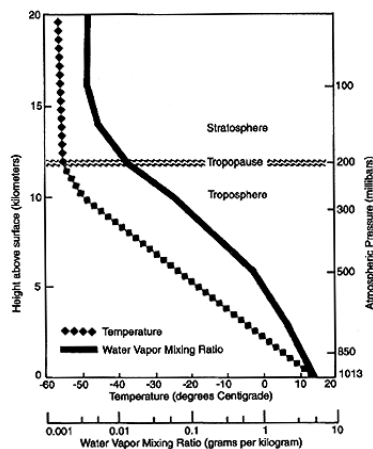
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13

The Atmosphere: water vapor

Close to the surface:

- from 1 g per kg of dry air in the very cold continental regions to 30 g per kg of dry air in hot maritime regions
- rare above 8000/9000 m



Q: Observed with which channel?

14

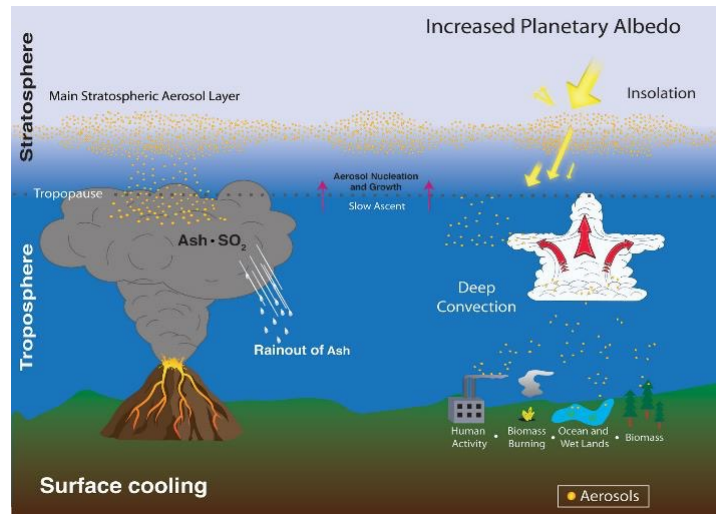
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14

The Atmosphere: aerosols

- **Hydrometeors:** water droplets and ice particles
- **Lithometeors:** vegetable and mineral wastes, bacteria, carbon particles,...
- light diffusion
- condensation nuclei

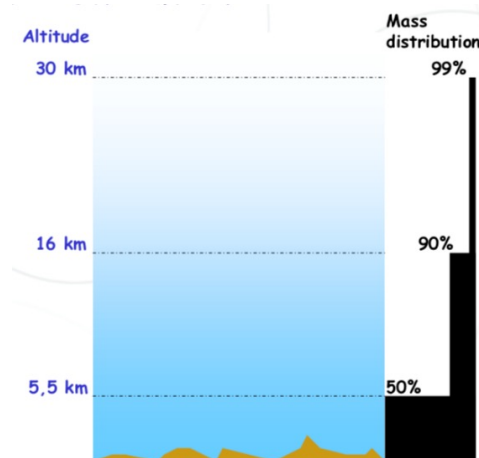
Q. Warming or cooling effect?



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15

The Atmosphere: vertical distribution (mass)



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16

16

The Atmosphere: vertical distribution

- **Mass:** The downward force of a unit volume of air:

$$F = \rho g$$

ρ : air density

g : gravitational acceleration

Integrating from the surface to the TOA $\rightarrow$ the atm. pressure on the Earth surface

$$p_s = \int_0^\infty \rho g dz$$

Assuming g equal to $g_0 = 9.807 \text{ ms}^{-2} \rightarrow p_s = mg_0$

where $m = \int_0^\infty \rho dz$ is the vertically integrated mass per unit area of the overlying air.

Q: The globally averaged surface pressure is 985 hPa. Estimate the mass of the atmosphere.

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17

The Atmosphere: vertical distribution

- Pressure p (and density ρ) decreases nearly exponentially with height:

$$p \simeq p_0 e^{-z/H}$$

where H is the *scale height*; p_0 is the pressure at some reference level, usually taken as sea level.

- In the lowest 100 km of the atm., $H \sim 7\text{-}8\text{ km}$.

$\rightarrow$

$$\ln \frac{p}{p_0} \simeq -\frac{z}{H}$$

E.g. $H=8\text{ km}$.

Q. 50% mass of the atm. is at approximately what height above sea level?

Q. 90% mass of the atm. is at approximately what height above sea level?

Q. 99% mass of the atm. is at approximately what height above sea level?

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18

Vertical distribution of pressure

- Q: How can we demonstrate the relationship?

$$p \simeq p_0 e^{-z/H}$$

Hint: the hydrostatic equation & the equation of state

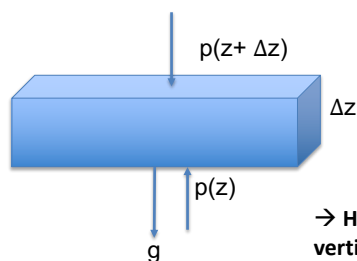
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19

19

The hydrostatic equation

- In an air parcel at rest



- Force downward: $-[p(z + \Delta z)\Delta x\Delta y + mg]$
- Force upward: $+[p(z)\Delta x\Delta y]$
- Equilibrium:

$$\frac{p(z + \Delta z) - p(z)}{\Delta z} \Delta x \Delta y \Delta z + mg = 0$$

→

$$\frac{\partial p}{\partial z} + \rho g = 0$$

→ **Hydrostatic balance equation:** balance between the vertical pressure gradient and gravity

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20

20

Vertical distribution of pressure

- We consider an isothermal atmosphere of temperature T_0 at rest.

The hydrostatic equation

$$\frac{\partial p}{\partial z} + \rho g = 0$$

The equation of state

$$p = R\rho T_0$$

→

$$p(z) = p_0 e^{-z/H}$$

→ Pressure and density decrease exponentially with height

$$\left. \begin{aligned} \frac{dp}{p} &= -\frac{g dz}{RT_0} = -\frac{dz}{H} \end{aligned} \right\} \text{where } H \text{ is the scale-height} = \frac{RT_0}{g}$$

- Suppose $T_0 = 0^\circ\text{C} \sim 273.15\text{K}$

$$\rightarrow H = \frac{287 \times 273.15}{9.807} \approx 8 \text{ (km)}$$

21

- Suppose $T_0 = -35^\circ\text{C} \sim 238.15\text{K}$

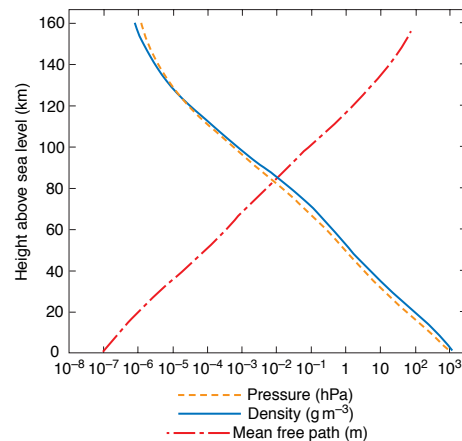
$$\rightarrow H = \frac{287 \times 238.15}{9.807} \approx 7 \text{ (km)}$$

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21

The Atmosphere (vertical structure)

- Density decreases with height in the same manner at pressure. **Q. Why?**
- the outermost reaches of the atm.: dominated by the lightest molecular species (H , H_2 , and He)
- Above $\sim 105\text{km}$, the **mean free path** exceeds 1m



Vertical profiles of pressure in units of hPa, density in units of kg m^{-3} , and mean free path (in meters) for the U.S. Standard Atmosphere.

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22

Exercise #1

Q. A person stands at the base of the Landmark 81 skyscraper.

- ✓ the air pressure at the ground level ($z_1 = 0 \text{ m}$) is **1013 hPa**
- ✓ the air density at the ground layer is **1.2 kg/m^3** .
- ✓ The person takes an elevator up to an observation deck at an altitude of **$z_2 = 400 \text{ m}$**

Assume the air density remains constant, what is the pressure at the observation desk?



Landmark 81 (HCM city)
461,2 m

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23

23

Exercise #2

A commercial passenger jet is cruising at an altitude of **10000 m**.

At this altitude, onboard sensors record the following environmental conditions:

- **Air Pressure (p):** 265 hPa
- **Air Temperature (T):** -50°C

Given the following constants:

- Gas constant for dry air (R_d): 287 J/kg.K
- Acceleration due to gravity (g): 9.81 m/s^2

- 1) Calculate the air density at cruising altitude
- 2) Calculate the pressure gradient at that altitude? *(then explain the significance for aviation)*

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24

24

Exercise #3

- Given the altitude of the summit of Mount Everest: ~8848 m
- Sea level pressure is 1013 hPa
- The scale height H is 8 km



Mount Everest

Q. Estimate the pressure at the summit of Mount Everest?

Q. What is the percentage of the atm.'s mass below this altitude?

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25

25

Practice #2 with Python

Vertical pressure profile in the atmosphere

$$p(z) = p_0 \cdot \exp(-z/H)$$

where $H=8000(\text{m})$; $p_0=1000.$; $z=0., 1000., \dots 9000.$; $dz=1000.$

1. Plot the vertical profile of pressure
2. Plot the analytical density at different atmospheric levels
3. Calculate (approximately) the density at different atmospheric levels by using forward, backward, central difference method, etc.
4. Compare the obtained simulation results with the analytical results.
5. If $dz=500$ (i.e. the levels are 0., 500., ..., 9000.). Compare the new simulation results with those obtained with $dz=1000.$

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26

Hint

The approximation of the derivative

$$\partial_x f^+ \approx \frac{f(x+dx) - f(x)}{dx} \quad \text{Forward differencing}$$

$$\partial_x f^- \approx \frac{f(x) - f(x-dx)}{dx} \quad \text{Backward differencing}$$

$$\partial_x f \approx \frac{f(x+dx) - f(x-dx)}{2dx} \quad \text{Centered differencing}$$

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